

Question ID: 2d2ab76b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$y = x^2 - 1$
 $y = 3$

When the equations above are graphed in the xy -plane, what are the coordinates (x, y) of the points of intersection of the two graphs?

Answer

- A. $(2,3)$
and $(-2,3)$
- B. $(2,4)$
and $(-2,4)$
- C. $(3,8)$
and $(-3,8)$
- D. $(\sqrt{2},3)$
and $(-\sqrt{2},3)$

Correct Answer: A

Rationale

Choice A is correct. The two equations form a system of equations, and the solutions to the system correspond to the points of intersection of the graphs. The solutions to the system can be found by substitution. Since the second equation gives $y = 3$, substituting 3 for y in the first equation gives $3 = x^2 - 1$. Adding 1 to both sides of the equation gives $4 = x^2$. Solving by taking the square root of both sides of the equation gives $x = \pm 2$. Since $y = 3$ for all values of x for the second equation, the solutions are $(2, 3)$ and $(-2, 3)$. Therefore, the points of intersection of the two graphs are $(2, 3)$ and $(-2, 3)$.

Choices B, C, and D are incorrect and may be the result of calculation errors.